

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

# ENVIRONMENTAL SCIENCE

## Paper 2 (Generated practice paper - Set 03)

Assessed: The living environment, Biological resources, Sustainability, Research methods

**Time allowed: 3 hours**

### Materials

For this paper you may use:

- a calculator.

### Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

### Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

### For Examiner's Use

| Question | Mark |
|----------|------|
| 1        |      |
| 2        |      |
| 3        |      |
| 4        |      |
| 5        |      |
| 6        |      |
| 7        |      |
| 8        |      |
| 9        |      |
| 10       |      |
| 11       |      |
| TOTAL    |      |

Answer all questions in the spaces provided.

0 1 . 1

A student measured the pH of a stream with a meter that reads to 0.01 pH units but had not been calibrated. Which statement is correct? Tick **one** box.

[1 mark]

A The readings are precise but may not be accurate

☐

B The readings are accurate but not precise

☐

C The readings are both accurate and precise

☐

D The readings are reliable because the meter is electronic

☐

0 1 . 2

Explain what is meant by an **anomalous result** and how a student should decide whether to include it.

[2 marks]

.....

.....

.....

.....

0 1 . 3

Explain why a preliminary study is carried out before the main investigation.

[2 marks]

.....

.....

.....

.....

5

0 2 . 1

State the Brundtland Commission definition of sustainable development.

[1 mark]

.....

.....

0 2 . 2

Explain the difference between a negative feedback mechanism and a positive feedback mechanism, giving **one** example of each in the climate system.

[4 marks]

.....

.....

.....

0 2 . 3

Explain why a diverse ecosystem is more likely to resist change than an agroecosystem.

[1 mark]

0 2 . 4

Evaluate the extent to which copying the principles of natural systems could make human activities more sustainable.

[9 marks]

Do not write outside the box

Extra space

15

03 . 1

Which of the following is an example of habitat destruction rather than a change in abiotic factors? Tick one box.

[1 mark]

- A Thermal pollution of a river by a power station
- B Flooding a valley to create a reservoir
- C Lowering the water table by groundwater abstraction
- D Acid mine drainage entering a stream

☐  
☐  
☐  
☐

03 . 2

Give **four** human activities that cause complete habitat destruction.

[4 marks]

1 .....

.....

2 .....

.....

3 .....

.....

4 .....

.....

5

04 . 1

Explain how the relationship between coral polyps and zooxanthellae allows a high biomass of life in nutrient-poor tropical seas.

[4 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

Do not write outside the box

|   |   |   |   |
|---|---|---|---|
| 0 | 4 | . | 2 |
|---|---|---|---|

Explain why growing coral reefs help to resist climate change.

**[2 marks]**

|   |   |   |   |
|---|---|---|---|
| 0 | 4 | . | 3 |
|---|---|---|---|

Discuss the importance of tropical coral reefs and evaluate the methods used to conserve them.

**[9 marks]**

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

Extra space

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

15

|   |   |
|---|---|
| 0 | 5 |
|---|---|

 . 

|   |
|---|
| 1 |
|---|

 Complete Table 1.

Do not write outside the

| Method              | Target species | One environmental disadvantage |  |
|---------------------|----------------|--------------------------------|--|
| Demersal trawling   |                |                                |  |
| Purse seining       |                |                                |  |
| Pelagic long-lining |                |                                |  |

**[2 marks]**

.....

.....

.....

.....

0 5 . 3

Explain what is meant by **ghost fishing** and describe **two** ways it can be reduced.

[2 marks]

.....

.....

.....

.....

10

0 6 . 1

Give **two** natural processes that are driven by solar energy.

[2 marks]

1 .....

.....

2 .....

.....

0 6 . 2

Explain why natural processes can take place at low temperatures.

[2 marks]

.....

.....

.....

.....

0 6 . 3

Give **one** human activity that uses high temperatures to achieve a result that natural systems achieve at low temperature.

[1 mark]

.....

.....

5

07 . 1

Give **two** habitats created by farming that support important wildlife communities.

[2 marks]

1 .....

2 .....

07 . 2

Explain how drainage and nutrient enrichment of farmland reduce biodiversity.

[2 marks]

.....

.....

.....

.....

07 . 3

Explain **one** problem that can be caused by introducing a biological control species.

[1 mark]

.....

.....

5

08

A new pond was dug in a nature reserve. Table 1 shows the vegetation recorded at intervals over 60 years.

Table 1

| Years after creation | Dominant vegetation                                | Water depth at centre / m |
|----------------------|----------------------------------------------------|---------------------------|
| 1                    | Single-celled algae, floating duckweed             | 2.0                       |
| 5                    | Submerged pondweeds; reeds at margins              | 1.8                       |
| 15                   | Reeds and reedmace over most of pond; water lilies | 1.0                       |
| 30                   | Marsh plants, sedges; willow and alder at edges    | 0.2                       |
| 60                   | Willow / alder carr; oak seedlings                 | 0 (soil waterlogged)      |

08 . 1

Name the type of sere shown in Table 1 and identify the pioneer species.

[2 marks]

08 . 2

Explain the changes shown in Table 1.

[5 marks]

08 . 3

Explain why a lithosere and a hydrosere in the same region end with the same climax community.

[2 marks]

08 . 4

Describe the early stages of a lithosere.

[1 mark]

10

**0 9** Table 1 shows some designated protected areas in the UK.

**0 9** . **1** Complete Table 1 to show the legislation under which each area is designated and its key feature.

**[4 marks]**

**Table 1**

| Designation                                | Designated under | Key feature |
|--------------------------------------------|------------------|-------------|
| Site of Special Scientific Interest (SSSI) |                  |             |
| Special Protection Area (SPA)              |                  |             |
| Ramsar site                                |                  |             |
| Marine Conservation Zone (MCZ)             |                  |             |

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

**0 9** . **2** Explain how designating a protected area protects habitats and species.

**[3 marks]**

.....

.....

.....

.....

.....

.....

.....

.....

**0 9** . **3** Explain why voluntary agri-environmental schemes are needed in addition to legislation.

**[3 marks]**

.....

.....

.....

.....

.....

.....

.....

.....

.....

10

**1 0** Table 1 shows the number of adult great crested newts caught in bottle traps in a pond over five years, and the number of ponds in the area in which the species was detected using environmental DNA (eDNA).

Table 1

| Year | Adults caught in traps | Ponds with newts detected by eDNA (out of 20) |
|------|------------------------|-----------------------------------------------|
| 2019 | 14                     | 6                                             |
| 2020 | 22                     | 9                                             |
| 2021 | 18                     | 11                                            |
| 2022 | 31                     | 13                                            |
| 2023 | 27                     | 15                                            |

**1 0** . **1** Calculate the percentage increase in the number of ponds where newts were detected between 2019 and 2023.

[2 marks]

Answer  %

**1 0** . **2** Explain how eDNA detects the presence of newts without catching them.

[2 marks]

.....

.....

10 . 3

Suggest why trap catches are a less reliable measure of population change than the eDNA survey.

[2 marks]

10 . 4

Evaluate the contribution of new technologies to ecological monitoring and wildlife conservation.

[9 marks]

Do not write outside the box

Extra space

15

Do not write outside the box

**Write an essay on one of the following topics.**

|   |   |   |   |
|---|---|---|---|
| 1 | 1 | . | 1 |
|---|---|---|---|

Discuss the methods that can be used to conserve biodiversity and evaluate their effectiveness.

**[25 marks]**

**OR**

**1 1 . 2**

Explain how the physical features of the Earth and the activities of living organisms have combined to create and maintain conditions suitable for life.

**[25 marks]**

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

[illegible]

[illegible]



**END OF QUESTIONS**